Mohamed Harzallah

AI Engineer / GenAI, AI Agents, LLMs, RAG

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PROFILE

AI Engineer with a strong technical background (Master's in Artificial Intelligence and Big Data, Université Paris Dauphine – PSL) and hands-on experience building LLM-based systems, agents and RAG pipelines in Python. Motivated by making AI systems reliable, safe and beneficial, I put a strong emphasis on rigorous evaluation, guardrails and automated testing. I enjoy turning ideas into working prototypes quickly, thrive in collaborative, fast-paced environments, and am eager to move into empirical AI research.

PROFESSIONAL EXPERIENCE

IRIS IT — AI Engineer (Final-Year Internship) · IRIS Agent *02/2025 – 08/2026*

Document intelligence platform for health insurance · Python, FastAPI, Next.js, MongoDB, Qdrant, LLM, Docker

- **Design and develop** autonomously a document intelligence platform (~20,000 lines, 3 containerized tiers), from OCR processing to human validation.
- **Orchestrate agents** with agno and build a multi-tenant hybrid RAG (dense search + BM25, RRF, LLM reranking, permission filtering).
- **Integrate LLMs via API** to extract structured data (constrained JSON outputs) across 6 document types and 48 business fields, including FR/AR handwritten documents, with deterministic fallbacks.
- **Implement guardrails** outside the LLM: HMAC confirmation tokens, identity binding and nonce-based input wrapping.
- **Detect anomalies** in an explainable way using 4 signals (fields, OCR text, dHash, embeddings) to distinguish duplicates, suspicions and unique documents.
- **Improve noisy OCR fields** by matching against the business database, using fuzzy similarity and field-type-specific normalization.
- **Evaluate quality** at every stage (per-field F1, recall@k, MRR, CER/WER, RAGAS-style faithfulness, ECE calibration) with 300+ automated tests.

Anypli — Final-Year Internship, Code Evaluation *03/2024 – 09/2024*

- **Develop a service** for isolated code execution (Docker) supporting 10+ languages, processing 1,000+ submissions per day in production.
- **Automate the grading** of 500+ exams, cutting grading time by 80%.

PROJECTS

TelecomPlus — Multi-Agent Assistance System | *LangGraph, LangChain*

- **Design a multi-agent architecture** that orchestrates a RAG agent to search the documentation and a SQL agent to query structured data based on the user's request.
- **Develop an LLM-as-a-Judge evaluation** mechanism to compare generated answers with expected ones, assign scores and produce justifications to analyze answer quality.

CV–Job Graph Matching | *Sentence-BERT, NetworkX, scikit-learn*

- **Develop a semantic matching pipeline** based on Sentence-BERT to encode CVs and job offers as vectors and measure their similarity to identify relevant matches.
- **Implement a match-prediction approach** combining graph algorithms with a Random Forest classifier.

SKILLS

GenAI & Agents: LLM APIs, prompting, context engineering, RAG, agents (agno, LangGraph, LangChain), orchestration, embeddings

Databases: Qdrant, MongoDB, PySpark, PostgreSQL, MySQL, Redis

NLP & Data: OCR, document extraction, Sentence-BERT, semantic matching, graph algorithms, data matching, CNN

Evaluation & Quality: automated testing, LLM-as-a-Judge

Engineering & Integration: Python, PyTorch, FastAPI, REST APIs

Tools & Methods: Docker, Git, GitHub, UML, Jira, Scrum, CRISP-DM, GCP

EDUCATION

Master's in Artificial Intelligence and Big Data — Université Paris Dauphine – PSL *2025 – 2026*

National engineering degree in Computer Science — Faculty of Sciences of Tunis *2021 – 2024*

First-Cycle University Diploma — IPEIM *2019 – 2021*

LANGUAGES

French: C1 · **English:** C1 · **Arabic:** Native